

Nishant Biyani

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SUMMARY

Ph.D. Candidate specializing in low-power Analog, RF, and Mixed-Signal IC Design. Experienced in PMICs, RF front ends, analog computing, and energy-efficient systems. Skilled in circuit and PCB design, schematic and layout implementation, and silicon validation. Seeking an internship in analog, RF, mixed-signal, and PMIC design, contributing across the IC development cycle from architecture and circuit design to silicon validation.

EDUCATION

Northeastern University, Boston, MA

July 2024 – May 2029 (Expected)

Ph.D. in Electrical Engineering, *Advisor: Aatmesh Shrivastava*

GPA: 4.0/4.0

- **Coursework:** Analog IC Design, VLSI Design, Power Management Integrated Circuits, RF IC design

Birla Institute of Technology Mesra, India

Aug. 2020 – May 2024

Bachelor of Technology in Electronics and Communication Engineering, (Gold Medalist)

GPA: 9.3/10

- **Coursework:** Digital System Design, Analog Circuits I & II, Electronic Devices

SKILLS

Programming & Scripting: Verilog, VHDL, C, Shell scripting, TCL, Python

IC Design & Tools: Cadence Virtuoso, Spectre, MATLAB, Simulink, HSPICE, Discovery Cluster

Analog and RF Subsystems: LNA, Mixer, VCO, BGR, Matching Networks, Comparators, DC-DC Converters, Low Dropout Regulator, Tapeout, Switched Capacitor Circuits, Energy Harvester, SoC Validation, System-Level Integration

Measurement and Lab Equipment: VNAs, Oscilloscopes, Function Generator, Logic Analyzers, Spectrum Analyzers, Signal Generator, Multimeter, Thermal Chambers

EXPERIENCE

Teaching Assistant, Northeastern University

Jan. 2026 – May 2026

- Guided students in analog CMOS circuit design and simulation using Cadence Virtuoso, MATLAB, and SPICE.
- Taught key analog building blocks, including op-amps, current mirrors, differential pairs, and active filters.
- Supervised layout implementation and performance evaluation through frequency, transient, and compensation analysis.

Summer Intern, EECS Lab Northeastern University

June 2023 – Aug. 2023

- Developed an automated RTL-to-GDS tapeout flow at the 65 nm TSMC technology node.
- Implemented flow using Cadence Virtuoso, Verilog, digital circuit design, and scripting.
- Enabled a fully automated chip tapeout pipeline, reducing manual effort and improving efficiency.

RESEARCH PROJECTS (DARPA & NSF GRANT), NORTHEASTERN UNIVERSITY

- Machine Learning Circuit Optimization Models for RF Energy Harvesting March 2026 – June 2026
- Ultra-low Power Analog Computing Machine Learning SoC for Seizure Detection Nov. 2024 – March 2026
- K-Means Clustering for Distance Calculation in Analog Computing July 2024 – Oct. 2024

ACADEMIC PROJECTS, NORTHEASTERN UNIVERSITY

- Designed a 45 nm CMOS bandgap reference with < 1 mV/°C temperature drift for precision biasing applications.
- Designed a 65 nm CMOS LDO regulator with 100 mV load regulation & sub-5 μ s transient recovery across a 1–100 mA load range.
- Designed a 45 nm CMOS DC-DC converter reducing **power consumption by 10x** compared to state-of-the-art.
- Designed a LNA in TSMC 65 nm improving **linearity by 1.5x** compared to state-of-the-art.

PUBLICATIONS

- **Nishant Biyani**, Hongling Liu, Utkarsh Kumar, and Aatmesh Shrivastava, “*Machine Learning Circuit Optimization Models for RF Energy Harvesting*,” **Accepted** at the 2026 IEEE 68th International Midwest Symposium on Circuits and Systems (MWSCAS), 2026.
- **Nishant Biyani**, Nikita Mirchandani, Yuqing Zhang, Kayland Harrison, Mostafa Abedi, Michael Drunov, Marvin Onabajo, and Aatmesh Shrivastava, “*A 265 Nano-Watt per Channel Analog Computing Machine Learning SoC for Seizure Detection*,” **Under Review** at the IEEE Journal of Solid-State Circuits (JSSC), 2026.
- **Nishant Biyani**, Dimitar Dimitrov, and Aatmesh Shrivastava, “*Ultra-low Power Current Mode Distance Calculation Circuit for K-Means Clustering*,” *2025 IEEE International Symposium on Circuits and Systems (ISCAS)*, London, United Kingdom, 2025, pp. 1–5. DOI:10.1109/ISCAS56072.2025.11043291